

Harshvardhan Singh Thakur

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Education

VIT Bhopal University

B.Tech in Computer Science; CGPA: 8.11

July 2023 – May 2027
Bhopal, Madhya Pradesh

Army Public School

Class XII; Percentage: 79.2%

Sagar, Madhya Pradesh

Army Public School

Class X; Percentage: 86%

Sagar, Madhya Pradesh

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional (Jul 2025)

IBM Generative AI Fundamentals (Jun 2025)

The Bits and Bytes of Computer Networking (Nov 2025)

Introduction to Machine Learning (NPTEL)(May 2025)

Projects

AgroMind: Real-Time Crop Auction Marketplace

- Architected a full-stack auction platform utilizing **Java Spring Boot, React, and PostgreSQL**. Engineered a high-throughput live bidding engine using an in-memory **Max-Heap (Priority Queue)** for instant concurrent bid resolution. Optimized data persistence with **Hibernate ORM** and enforced transactional integrity using **PL/SQL triggers**, with core services tested via **JUnit and Mockito**.

VidyaDrishti: Rural Education Analytics Tool

- Developed an EdTech platform focused on optimizing learning in village schools. Engineered a custom **'Focus Score' algorithm** to track and quantify student engagement. Optimized backend **REST APIs** and database schemas to ensure reliable data persistence (focus metrics) in low-bandwidth environments.

Medicine Recommendation System

- Developed a system that predicts diseases based on symptoms and recommends appropriate medicines, precautions, diet, and workout plans. Implemented real-time symptom matching and machine learning for accurate predictions.
- Stack:** Python, Pandas, Scikit-learn, Fuzzy String Matching, Random Forest

Collaborative Text Editor

- Engineered a real-time collaborative text editor utilizing STOMP over WebSockets for concurrent, low-latency document synchronization across multiple clients.
- Integrated GenAI capabilities via the Google Gemini API and Spring Boot RestClient to provide seamless, contextual autocomplete suggestions directly within the UI.
- Stack:** Java, Spring Boot, React, STOMP WebSockets, Google Gemini API

Technical Skills

Programming Languages: Java, JavaScript, Python, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Spring Boot, Spring MVC, Spring Core, Spring Data JPA, Hibernate, Node.js, Express.js

Databases: MySQL, MongoDB

Tools: Git, GitHub, Maven, Postman, Eclipse IDE, VS Code, MySQL Workbench, AWS, Vercel, Render

Core CS: Data Structures & Algorithms, Object-Oriented Programming,, REST APIs, MVC Architecture, Dependency Injection, IoC, ORM, JPA

Achievements & Responsibilities

Smart India Hackathon (SIH) - Internal Round Qualifier

- Among the top 75 teams selected in University Internal Round to represent the institute for the **Smart India Hackathon**.

Hackathon Participation

- Active participant in the **Hack-n-Go MongoDB Hackathon** and **ShikshaLokam Hackathon**. Built functional prototypes under strict deadlines, utilizing MongoDB and rapid application development strategies.

NSS Volunteer, VIT Bhopal (July 2025)

- Contributed to community welfare initiatives as an active volunteer. Successfully coordinated a large-scale plantation drive, demonstrating teamwork, event management skills, and social responsibility.